FINAL YEAR PROJECT · INDUSTRY SYMPOSIUM 2026

IntelliQuery

AI-Powered Conversational Business Intelligence Platform

ENTERPRISE VALUE PROPOSITIONS

NL → SQL Agent

Model: LangGraph + Llama-3.3-70B via Groq

Context: 23 curated few-shot examples

Memory: 10-turn multi-turn conversation

Output: Deterministic, reproducible SQL

Zero SQL expertise needed

12 Auto-Rendered Charts

Types: Bar, Line, Pie, Treemap, Radar, Funnel…

Selection: LLM payload drives chart type choice

Layout: Drag-and-drop React Grid dashboard

Export: PNG / PDF from any chart or view

No configuration required

RAG over Company Docs

Formats: PDF, DOCX, CSV, TXT, MD

Search: ChromaDB + FastEmbed vector store

Fallback: DuckDuckGo web at confidence <0.30

Answers: Source-cited with document + page ref

Cited · Grounded · Accurate

Production-Grade Security

Injection: Regex-based SQL injection detection

Blocking: DROP / DELETE / ALTER / TRUNCATE

Validation: PostgreSQL EXPLAIN pre-execution

Recovery: 3× auto-retry with error classification

Enterprise-ready guardrails

SYSTEM ARCHITECTURE — END-TO-END DATA FLOW

USER INTERFACE LAYER

React 19 + Tailwind CSS + Framer Motion

NL Chat · Drag-and-Drop Dashboard · Manual Chart Builder · RAG Upload

React 19 · Tailwind CSS · Recharts · React Grid Layout · Axios

HTTP / REST (AXIOS)

API GATEWAY

FastAPI Backend + JWT Authentication

Session Management · Query Router · CORS · Pydantic v2 Validation

FastAPI · Uvicorn · PyJWT · Pydantic v2

SESSION CACHE

Redis

7-day TTL · Dashboard state · Conversation history · In-memory fallback

Redis · 7-day TTL · In-mem fallback

MULTI-LAYER SECURITY GUARDRAILS — CRITICAL PRODUCTION LAYER

SQL Injection · Keyword Blocking (DROP/DELETE/ALTER) · Syntax Validation · EXPLAIN Pre-Check · 3× Auto-Retry

Read-Only Enforced · Regex Pattern Scan · Parenthesis Balance · Business Logic Checks · Error Classification

SCHEMA INTELLIGENCE

Schema Inspector

Semantic table selection · 40+ keyword→table mappings · FK detection · Business context injection

Semantic Scoring · Fuzzy Match

AI ORCHESTRATION CORE

↺ 3× Auto-Retry

LangGraph · Llama 3.3 · 70B via Groq

StateGraph · Conditional routing · Few-shot semantic retrieval · Temperature: 0 (deterministic)

LangGraph · LangChain · Groq API · FastEmbed

QUERY PLANNING

SQL Planner + Validator

Regex cleaning · Syntax check · Business logic validation · EXPLAIN pre-run

FastMCP · stdio transport

RELATIONAL DATA STORE

PostgreSQL — Northwind Dataset

Live SQL execution via psycopg2 · 14 tables · 3,000+ rows · Decimal→float, date→ISO

psycopg2 · SELECT-only · 14 tables

VECTOR KNOWLEDGE STORE

ChromaDB + FastEmbed · RAG Pipeline

800-char chunks · 120-char overlap · Confidence threshold 0.30 · DuckDuckGo fallback

ChromaDB · FastEmbed · DuckDuckGo · Source Citations

INTELLIGENT OUTPUT LAYER

Chart Recommender → Recharts Dashboard

Keyword + data-structure analysis selects from 12 chart types · Cross-filter propagation · PNG/PDF export · Auto-generated starter dashboard on login

Bar · Line · Pie · Treemap · Radar · Funnel · Scatter · KPI Card · Cross-Filter · PNG Export

SYSTEM PERFORMANCE & PRODUCTION READINESS

23

Few-Shot Examples

Curated Northwind SQL examples for semantic retrieval

40+

Semantic Mappings

Keyword-to-table scoring in Schema Inspector

5

Error Categories

Syntax · Permission · Connection · Logic · Timeout — each with auto-recovery

7d

Redis Persistence

Sessions · Dashboard layouts · Conversation history · In-memory fallback

Supervisor: **Abeera Tariq**

IBA SMCS · School of Mathematics and Computer Science